

GCC Lab المفتبر الفليبي	Gulf Laboratory Technical Services
	Equipment Procedure for Determination of Total Acid Number By Metrohm

Equipment Number: EQ-79 & EQ-80
Serial Number: 1905001043998 & 1814001020628



1. PURPOSE

This standard operating procedure is intended to provide users the guide on the operation and maintenance of Metrohm 907 Titrando with 814 USB Sample Processor Potentiometer

2. SCOPE

This standard operating procedure is employed in the Gulf Laboratory Technical Services Co. primarily for acid number test, pH determination, peroxide determination in mineral oil and reagents standardization by tiamo automation system titration.

3. TERMS AND DEFINITIONS

3.1.1 Tiamo – is a control data base software for titrators and dosing devices

3.1.2 907 Titrando – is the metrohm exchange unit system that equipped with data chip

3.1.3 814 USB Sample Processor – is a component of the modular titrando system.

4. RESPONSIBILITIES

4.1 Use, care and maintenance of the Metrohm 907 Titrando with 814 USB Sample Processor Potentiometer is directly assigned to the end user who is usually the Sr. laboratory chemist. This responsibility may be shared to other qualified user(s) upon the consent of the Sr. laboratory chemist.

Doc.: EOP [6]-04-L	Version No.00	Issue Date: 1/11/2023	Revision No: 00	Date:1/11/2023
Prepared by: Quality Manager	Reviewed by: Technical Manager	Approved by: General Manager	Page 1 of 5	

GCC Lab المفتبر الفليبي	Gulf Laboratory Technical Services
	Equipment Procedure for Determination of Total Acid Number By Metrohm

4.2 Application of this responsibility is based on the Memorandum Receipt and further supported by a Memorandum of Authorization both signed by the General Manager.

5. SAFETY PRECAUTIONS

5.1 Use a proper electrical outlet of correct voltage and current handling capacity.

5.2 Disconnect from the power supply prior to maintenance and servicing.

5.3 Ensure that the operating switch operates properly.

6. PRINCIPLES OF OPERATION

6.1 The Titrand System

The titrand is the core of the modular titrand system ("Stand alone" titrator). Operation is carried out by a personal computer controller with a corresponding software tiamo, a network titration data system. It has the following features, dosing, MSB Connectors Connectors and measuring interface with one high ohm measuring input Ind for pH, redox or ISE electrode, for reference electrode and other etc. for complete titration automation system.

7. OPERATIONAL DETAILS

7.1 General Instructions

7.1.1 Turn on the entire device system.

7.1.2 Start the software by clicking the Tiamo icon twice in computer monitor.

7.1.3 Tiamo 2.2 configuration window will appear with four programs:

Workplace
Database
Method
Configuration
Manual

7.2 **Workplace** – To start single / series determination and create sample tables

7.2.1 Open the workplace window

7.2.2 Open the run window

7.2.3 Open the edit line sample table by clicking the method.

Doc.: EOP [6]-04-L	Version No.00	Issue Date: 1/11/2023	Revision No: 00	Date:1/11/2023
Prepared by: Quality Manager	Reviewed by: Technical Manager	Approved by: General Manager	Page 2 of 5	

GCC Lab المفتبر الفليبي	Gulf Laboratory Technical Services
	Equipment Procedure for Determination of Total Acid Number By Metrohm

7.2.4 Choose the method of analysis in the list table.

7.2.5 Entry the data like sample position number from 814 USB Processor, sample ID and sample size and click apply.

7.2.6 Click next (►) symbol if more samples to be analyze and apply and finally click the close icon to close the table.

7.2.7 Click start icon at the top to start up the instrument operation.

7.3 Database – To evaluate and reprocess the data results and method templates.

7.3.1 Open the database window

7.3.2 From determination overview window, select the sample ID of finished analysis to show the sample potentiogram, complete information and results overview.

7.3.3 To reprocess the potentiogram, open the curves 1 and click the reprocess icon, open the curve evaluation and edit by selecting list of command icons, save and recalculate.

7.3.4 To print the report. Just click file icon and click print report.

7.4 Method – To create new methods or manage methods of operation

7.4.1 Open the method window

7.4.2 Select the kind of track needed to perform by 814 USB Sample Processor and 888 Tirando (Normal Track, Series Track, Exit Track and etc.)

7.4.3 Click the command icon and select which command to be perform and from the list of command select a specific operation to be done.

7.4.5 Continue same process until a complete track of command needed for the method is finished.

7.4.6 An end series track should be made to complete the method operation and save the method file name of the operation.

Note: For more detailed procedure refer to Tiamo Tutorial Manual.

7.5 Configuration – To configure devices, solutions, sensors and common variables

7.5.A Device Window - 888 Titando and 814 USB Sample Processor are already installed

Doc.: EOP [6]-04-L	Version No.00	Issue Date: 1/11/2023	Revision No: 00	Date:1/11/2023
Prepared by: Quality Manager	Reviewed by: Technical Manager	Approved by: General Manager	Page 3 of 5	

	Gulf Laboratory Technical Services
	Equipment Procedure for Determination of Total Acid Number By Metrohm

7.5.B Sensors Window – shows the available electrode type currently installed or used for selected method with corresponding microchips information.

7.5.C Titrant Solution Table – shows the data information of titrant used by 888 Titrandos like solution name, concentration, titer and etc. which detected by sensor. Can be edited.

7.5. D Common Variables Window – Define the variables used in the method calculation formula. Can be edited.

7.6 **Manual Window** – For manual dosing, preparing exchange unit, stirring and automation functions.

8.0 Calibration and Verification

8.1 Calibrate electrode using 3 different pH buffers (4, 7 and 9).

9.0 Maintenance

9.1 Replacing the dosing/exchange unit

9.1.1 Transfer the burette tip and the electrode of exchange unit system into its unit holder.

9.1.2 Unplug the electrode cable from the 888 Titrandos cable inlet.

9.1.3 Slowly pull forward the exchange unit system to release from the Titrandos

9.1.4 Install the new exchange unit system into the Titrandos. Make sure that it's the unit required by the method; sensor will read the chips information.

9.2 Preparing the new exchange unit

9.2.1 Connect the electrode cable to the Titrandos cable inlet at the back of the unit.

9.2.2 Keep the burette tip directed into the vessel.

9.2.3 Open the manual control window and select dosing device and open prepare window.

9.2.4 Click start until the cylinder and the tubing of exchange unit is air bubble-free.

9.2.5 Close the manual control and transfer the burette tip and electrode to 814 USB Sample Processor unit holders.

Doc.: EOP [6]-04-L	Version No.00	Issue Date: 1/11/2023	Revision No: 00	Date:1/11/2023
Prepared by: Quality Manager	Reviewed by: Technical Manager	Approved by: General Manager	Page 4 of 5	

GCC Lab المفتبر الفليبي	Gulf Laboratory Technical Services
	Equipment Procedure for Determination of Total Acid Number By Metrohm

9.3 Electrode Maintenance

- 9.3.1 Clean the glass electrode (Note 7) at frequent intervals (not less than once every week during continual use) by immersing in cold chromic acid cleaning solution or in other equipment cleaning solutions.
- 9.3.2 Drain the electrode at least once a month and refill with fresh KCl electrolyte as far as the filling hole. As certain that crystallized KCl is present. Maintain the electrolyte level in the electrode above that of the liquid in the titration beaker or vessel at all times. When not in use, immerse the lower halves of the electrodes in water. Do not allow them to remain immersed in titration solvent for any appreciable period of time between titrations. While the electrodes are not extremely fragile, handle them carefully at all times.
- 9.3.3 Before and after using, wipe the glass electrode thoroughly with a clean cloth, or a soft absorbent tissue, and rinse with water. Wipe the electrode with a cloth or tissue, carefully remove the ground-glass-sleeve and thoroughly wipe both ground surfaces. Replace the sleeve loosely and allow a few drops of electrolyte to drain through to flush the ground-glass joint (**Warning**—Causes severe burns.). Wet the ground surfaces thoroughly with electrolyte, set the sleeve firmly in place, and rinse the electrode with water. Prior to each titration, soak the prepared electrodes in water for at least 5 min immediately before use, and touch the tips of the electrodes with a dry cloth or tissue to remove the excess of water.

10.0 REFERENCE

- 10.1 IEC 62021-1 (Insulating liquids- determination of acidity- part 1: automatic potentiometric titration).
- 10.2
10.2 Tiamo Tutorial Manual

Doc.: EOP [6]-04-L	Version No.00	Issue Date: 1/11/2023	Revision No: 00	Date:1/11/2023
Prepared by: Quality Manager	Reviewed by: Technical Manager	Approved by: General Manager	Page 5 of 5	